

CYBERSECURITY: PREVENTION AND INTERVENTION · LEA.D8 · AEC

Class 1

Introduction to Preventative Monitoring

420-1C1-DW · 75 hours · ponderation 2-3-1 · competency EZ01
Monday 14 September 2026 · online · Dawson College
Asen Grozdanov · Cyber.SoHo



Educational Hub · From the Industry to the Classroom

What this hour covers

Four lessons, one continuing case study, four deliverables you will hand in later.

1

Course outline

What the course code, the hours and the competency actually oblige you to do.

2

Schedule and evaluation

Nineteen classes, nine graded items, two rules that decide whether you pass.

3

What monitoring is

A precise definition, the seven-stage pipeline, and what a baseline is.

4

Lab environment

The isolated laboratory every exercise runs in, and why it must stay isolated.

The continuing case study. Atelier Nord-Ouest is a fourteen-person architectural drafting and printing studio in Verdun, Montréal. Every example tonight, and all three laboratories, use the same network. Nothing has to be re-established later in the term.

Everything about it is fictional, as section 6.7 of the Cyber.SoHo notice requires.



The rule that governs everything else

Authorisation, in writing, from the owner.

Every technique in this course is lawful when the owner of a system has authorised you, and a criminal offence when that authorisation is absent. The purpose protects nobody; the authorisation does.

If you are not certain that you have authorisation, you do not have it.

s. 342.1

Unauthorised use of a computer — up to 10 years

s. 342.2

Possession of a device, including software — up to 2 years

s. 430(1.1)

Mischief in relation to computer data — up to 10 years

s. 184

Interception of private communications — up to 5 years

Read before the first laboratory: Cyber.SoHo Legal Notice, Terms of Use and Acceptable Use Policy, v2.0 — section 7.2 applies to this course. Sign Annex A.

<https://cyber-soho.github.io/prevmon/00.01.%20cyber.soho%20-%20legal.html>



LESSON 1 OF 4

1.1 Course outline

What the course code, the hours, the credits and the competency oblige you to do — read as a specification, not as an announcement.



Reading the course code and the ponderation

420

Computer science and
information technology

1C1

Term 1, course 1 of the
cybersecurity stream

DW

Dawson College,
the issuing institution

2 - 3 - 1

Theory · laboratory ·
personal study, per week

2 ⅔

Credits, calculated from
the ponderation

What 2-3-1 actually obliges you to do

The third number is not a suggestion. One hour of personal study per week is the minimum the credit calculation assumes; it is already counted in your 2⅔ credits, which means it is already counted in the workload the college expects of you.

75 hours total = **30 theory + 45 laboratory** · compressed into six weeks rather than fifteen.



Competency EZ01

This course addresses EZ01 in its entirety. No other course in the programme covers it.

Statement of the competency

Monitor a computer network from a cybersecurity perspective.

Achievement context

For an entire documented network of various platforms · ongoing monitoring · under supervision · in collaboration with technicians and users.

Based on

Policies, norms and standards · descriptive infrastructure documents.

With the help of

Technical manuals · equipment, tools and software · a scripting language · network protocols.

Why the wording matters. “Under supervision” and “in collaboration” are graded behaviours, not context. An examiner can mark you down for acting alone on something that required a colleague.



The five elements of EZ01

Each element is taught by a named topic and assessed by a named laboratory. Nothing is left implicit.

1

Distinguish the concepts linked to cybersecurity

Confidentiality, integrity, availability · types of attack · defence concepts · security versus performance

Topic 1

2

Analyse the topology of the network

Organisational and operational constraints · surveillance needs · prioritisation against resources

Topic 2

3

Install and configure surveillance tools

Tool selection · predefined suspicious behaviours · updates · reference database · alerts

Topic 3

4

Detect and diagnose anomalies

Correct use of the tools · methodical analysis of traffic, files and event logs · accurate diagnosis

Topics 4 – 5

5

Optimise surveillance tools

Update the reference databases · limit impact on performance · widen coverage · improve accuracy

Topic 6



The six topics, in the order you will meet them

1

Foundations of cybersecurity

CIA triad · attack types · defence concepts · security versus performance issues

2

Understanding the network

Components · documentation and topology · constraints · needs · prioritisation

3

Monitoring tools

Categories · selection · installation and updates · the baseline · alert configuration

4

Collecting and analysing data

Event logs · traffic capture · network files · filtering · normal versus abnormal

5

Detecting and diagnosing anomalies

What an anomaly is · triage · step-by-step analysis · diagnosis · communication

6

Optimising the setup

Updating baselines · performance impact · coverage · accuracy · recommendations

Topics 1 to 3 are examined at the midterm on 1 October. **All six are examined at the final on 19 October.**



Which topic and which laboratory cover each element

Filled circle = taught or assessed here. Open circle = reinforced here.

Competency EZ01 — coverage by topic and by laboratory (420-1C1-DW, Fall 2026)

Filled circle = element is taught or assessed here. Open circle = element is reinforced.

	T1	T2	T3	T4	T5	T6	Lab 1	Lab 2	Lab 3
1 · Distinguish cybersecurity concepts	●			○			○		
2 · Analyse the topology of the network		●	○				●		
3 · Install and configure surveillance tools			●			○		●	
4 · Detect and diagnose anomalies				●	●			○	●
5 · Optimise surveillance tools						●			●

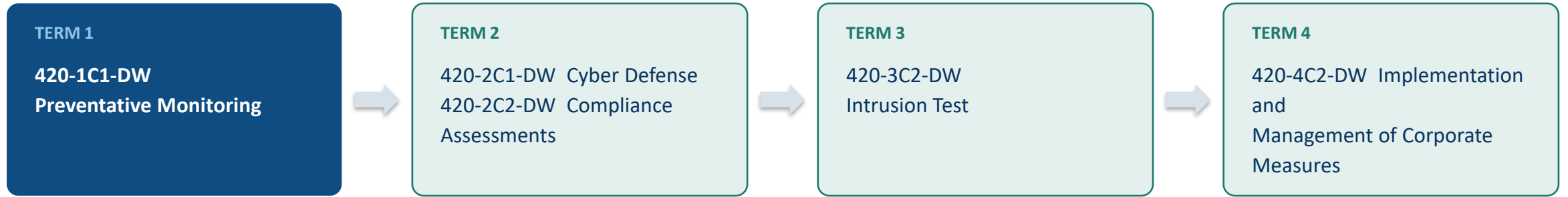
T1 Foundations · T2 Network understanding · T3 Monitoring tools · T4 Data collection and analysis · T5 Anomaly detection · T6 Optimisation
 Lab 1 = 25 Sept · Lab 2 = 3 Oct · Lab 3 = 16 Oct · each worth 5 % · together they form the single Integrative Project

Every element is covered by at least one topic and assessed by at least one laboratory. A row with no filled circle would be a defect in the course, not in your revision.



Where this course leads

Preventative Monitoring is a prerequisite for four later courses. Failing it stops the programme, not just the term.



What the later courses assume you can already do

Read a topology and say what must be watched first · install a collector and know what it is collecting · tell a security problem from a performance problem · write a finding somebody else can act on.

Intrusion Test in Term 3 is the strictest course in the programme. It assumes the discipline you build here.



Atelier Nord-Ouest — the network you will monitor

Fourteen people, one small network, three laboratories. Fictional throughout.

Router / firewall

the only path to the internet

Windows file server

holds the client drawings

Linux backup server

the last line of recovery

Managed switch

provides the mirror port

Two access points

unmanaged entry point

Ten workstations

where malicious files arrive

Large-format printer

embedded, rarely patched

Camera recorder

records identifiable people

Deliverable D1 — the monitoring scope register. For each asset: why it is monitored, which elements of EZ01 it lets you demonstrate, and what evidence you hand in.

Due with Laboratory 1 on Friday 25 September. Tonight's program builds the first version for you.



Building the scope register, live

Thirty lines of Python. Each asset is added, and the coverage of EZ01 is recalculated after every one.

EZ01 coverage, built live

1 · Cybersecurity concepts	covered
2 · Analyse network topology	covered
3 · Install and configure tools	covered
4 · Detect and diagnose anomalies	covered
5 · Optimise surveillance tools	covered
5 of 5 elements covered	

What the program does

Walks the eight assets, records which elements each one demonstrates, reports any element nothing covers, and writes the register to a CSV file.

Why it matters

A scope that leaves an element uncovered is a scope that cannot be assessed. The program turns that from a judgement call into a check you can run.

Result: **5 of 5 elements covered, 0 gaps.**



Watch and read for Lesson 1.1

Watch before the next class — each under 15 minutes

A day in the life of a SOC analyst · 12 min <https://www.youtube.com/watch?v=r6uGbvUdyVw>

IBM — what a SIEM is, in four minutes <https://www.youtube.com/watch?v=9RfsRn7m7OE>

Institutional documents

- Institutional Student Evaluation Policy (ISEP) — your rights and obligations on evaluation, grade review and grievance.
- Dawson College — programme and departmental information.
- Cyber.SoHo Legal Notice v2.0 — section 7.2 governs this course.

Library account — set it up this week

A free account with the Bibliothèque et Archives nationales du Québec (BANQ) gives you O'Reilly and LinkedIn Learning at no cost.

cap.banq.qc.ca/inscription?locale=en

Every address used tonight is collected at the end of the lecture page, with a note on what it is for. **Check any link not on that list with VirusTotal before you open it.**



LESSON 2 OF 4

1.2 Schedule and evaluation

Nineteen classes, nine graded items and two rules that decide, between them, whether you pass.



The term at a glance

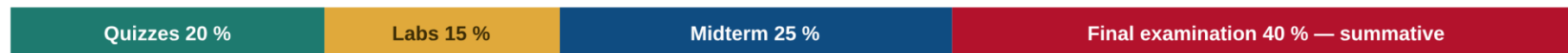
14 September to 19 October 2026. Sand = online · navy = written in class · crimson = statutory closure.

420-1C1-DW — term calendar, 14 September to 19 October 2026

Sand = class delivered online · Navy = written in class · Crimson = statutory closure, no class



Weight of each component, drawn to scale



Pass condition

Final examination $\geq 60\%$. A result below 60 % fails the course and becomes the recorded course grade.

Weighted total $\geq 60\%$. Both conditions must be satisfied.

Late work: -10% per day, refused after 7 days. Literacy: up to -10% for grammar, spelling, punctuation and syntax.

Two Mondays are lost: **5 October, provincial election day** and **12 October, Thanksgiving**. Nothing is rescheduled — the content is absorbed into the surrounding classes.



How the classes are delivered

19

classes in total

17

delivered online

2written in class:
1 October and 19 October**2**statutory closures:
5 and 12 October**6**

weeks, start to finish

DAY	TIME	MODE	WHAT NORMALLY HAPPENS
Monday	18:00 – 22:00	Online	New material, or a review session before an examination
Thursday	18:00 – 22:00	Online	New material · both examinations fall on a Thursday or Monday, in class
Friday	18:00 – 22:00	Online	New material · the three laboratories are set on Fridays or a Saturday
Saturday	14:00 – 18:00	Online	New material · the cumulative quizzes are usually set here



The nine graded items

Everything that counts towards your grade, with its date and its weight. There is nothing else.

Four cumulative quizzes

19 Sept · 26 Sept · 10 Oct · 15 Oct

5 % each

20 %

Three laboratories

25 Sept · 3 Oct · 16 Oct

5 % each

15 %

Midterm examination

1 October, in class

Topics 1 to 3

25 %

Final examination

19 October, in class

Summative · Topics 1 to 6

40 %

Nothing is optional, and nothing else is graded.

Total 100 %



Two rules decide whether you pass

Rule 1

The final examination must reach 60 %.

It is the summative assessment. A result below 60 % fails the course, and that result becomes the grade recorded for the course — whatever your other marks were.

Rule 2

The weighted total must reach 60 %.

The ordinary passing mark, applied to all nine items together. Passing the final examination does not excuse a weighted total below 60 %.

Both conditions must be satisfied. Satisfying one is not passing the course.

The passing grade for every individual assessment is also 60 %.



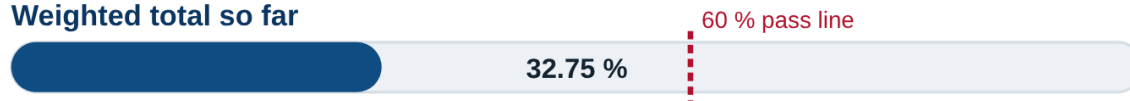
What the arithmetic actually demands

A worked example: a student who has written five of the nine items and wants 75 % overall.

Weighting accumulated, and the pass test



Weighted total so far



Rule 1 — final examination ≥ 60 % : needs 77.5 % for the 75 % target

Rule 2 — weighted total ≥ 60 % : 32.75 earned, 55 % still available

The worked numbers

Quiz 1 80 %, Quiz 2 72 %, Lab 1 85 %, Midterm 68 %, Lab 2 78 %.

Earned so far: **32.75 points of 100**

Still to be written: **55 % of the course**

Needed on the final for 75 % overall: **77.5 %**

The programme calculates this from the plan, not from a guess. Any student can run it with their own marks.



Why every quiz is cumulative

Each quiz covers all material presented up to its date, not only the week just finished.

Quiz 1 · 19 Sept

Topic 1 in full: the CIA triad, attack types, defence concepts, security versus performance.

Quiz 2 · 26 Sept

Everything in Quiz 1, plus Topic 2 and the opening of Topic 3.

Quiz 3 · 10 Oct

Everything so far, plus Topic 4 and the start of Topic 5.

Quiz 4 · 15 Oct

Everything so far, including the whole of Topic 5.

This is a deliberate design, not an inconvenience

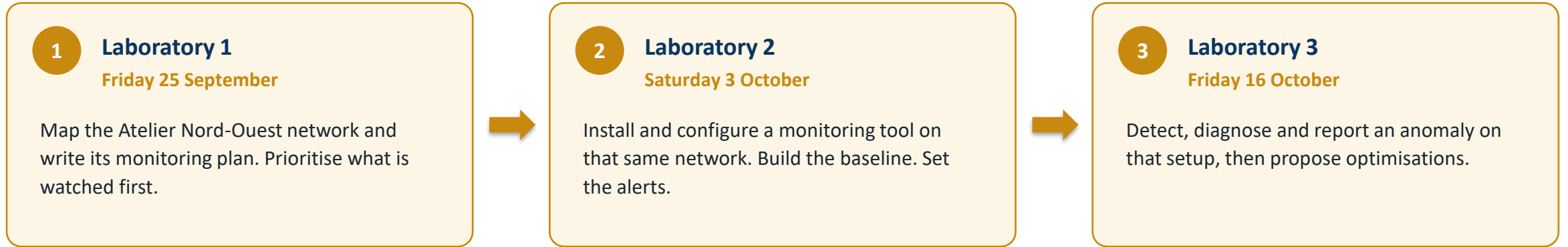
Retrieving material repeatedly, spaced over weeks, produces far more durable knowledge than reading it repeatedly. The effect is one of the best-established findings in the study of learning, and the quiz schedule exists to force it.

The practical consequence: **Topic 1 is examinable on four separate dates, and again at both examinations. It is the cheapest topic to master and the most expensive to neglect.**



The three laboratories are one project

Each is graded separately at 5 %, but each builds directly on the one before it.



What this means for you in practice

Work handed in late for Laboratory 1 delays Laboratory 2, because Laboratory 2 needs the network you mapped. Assembling the three parts into one document at the end is optional and receives written feedback only — it does not change your grade.



The rules around the marks

Late submissions

–10 % for each day late. Nothing is accepted more than 7 days after the due date.

Missed tests and quizzes

A valid reason can be accommodated with a make-up. Under 5 days, no medical certificate. Over 5 days, a certificate is required.

Literacy

Up to 10 % may be deducted for grammar, spelling, punctuation and syntax. A piece of work may be sent back to be rewritten.

Academic integrity

Cheating and plagiarism fail the assessment or the whole evaluation component, and may fail the course. Every case is reported to the Dean.

Conduct

The Student Code of Conduct applies in every class, online included. Electronic dictionaries and telephones are prohibited during tests.

Attendance

Governed by the ISEP. Classes are not recorded for students who miss them, and recording a session without written permission is prohibited.

All of this comes from the ISEP and the course outline. **Where a Cyber.SoHo rule and a College rule overlap, comply with both; where they conflict, the stricter one governs.**



Watch and do for Lesson 1.2

Watch before the next class — each under 15 minutes

Active recall and spaced repetition, for beginners · 8 min <https://www.youtube.com/watch?v=CBEO9TG0Q7U>

A three-minute summary of the same technique <https://www.youtube.com/watch?v=03GGA-mkJ34>

Do these three things before Thursday 17 September

- Put all nine dated items into whatever calendar you actually look at, with an alert three days before each.
- Block the two in-class dates — 1 October and 19 October — including travel time.
- Open a free BANQ account so O'Reilly and LinkedIn Learning are available when you need them.

Deliverable D2 — your personal delivery and study plan — is handed in with Laboratory 1 on 25 September.



LESSON 3 OF 4

1.3 What preventative monitoring is

A precise definition, the seven stages between a raw observation and a usable report, and the one statistic the whole course rests on.



The definition, clause by clause

The ongoing observation of a computer network in order to detect and report problems before they cause damage.

ongoing

Not a one-off audit. A setup that runs for a day tells you about that day and nothing else.

observation

Watching, not changing. Monitoring collects evidence; it does not remediate. Changing a live system is a separate, authorised act.

of a computer network

The whole documented network of mixed platforms, as the competency requires — not a single machine.

detect and report

Two obligations, not one. An alert nobody reads and cannot act on has no value.

before they cause damage

The word that makes it preventative. The measure of success is how early you see it, not how completely you describe it afterwards.

A common confusion to settle now: monitoring is a detective control, not a preventive one. It prevents damage by shortening the time to detection — it does not stop the event.



The CIA triad — what you are protecting

Confidentiality, Integrity, Availability. Every security question in this programme reduces to which of these three is at risk.

C Confidentiality

Only the people entitled to see the information can see it.

At Atelier Nord-Ouest: a client's drawings must not leave the file server.

Monitored by: who authenticated, who opened which share, what left the network.

I Integrity

The information is what it is supposed to be, and has not been altered.

At Atelier Nord-Ouest: a drawing must not come back from backup altered.

Monitored by: file hashes, change events, configuration drift.

A Availability

The information and the service are there when they are needed.

At Atelier Nord-Ouest: the printer must work on the afternoon a submission is due.

Monitored by: uptime, response time, queue depth, error rates.

One event often threatens more than one letter. Ransomware on the file server destroys availability immediately, destroys integrity because the files are altered, and usually destroys confidentiality as well, because the data is copied out before it is encrypted.



Security issue or performance issue?

Element 1 of EZ01 asks for a clear distinction between the two. The same symptom can be either.

Performance issue

A resource is insufficient or badly configured. Nobody intended harm.
The fix is capacity, configuration or scheduling.

Security issue

Someone is doing something they are not entitled to do, or a control has failed. The fix involves authorisation, containment and an incident record.

OBSERVATION	COULD BE A PERFORMANCE ISSUE	COULD BE A SECURITY ISSUE
The file server is slow at 09:00 every weekday	Everyone arrives and opens files at once — capacity	Credential-stuffing against the logon service — attack
The backup job did not finish last night	The volume filled up — capacity	The backup service was stopped deliberately — tampering
Traffic to one external address has tripled	A large software update is downloading — expected	Data is being copied out of the network — exfiltration

The observation never tells you which. Only the surrounding evidence does — which is why the diagnosis stage exists and why monitoring collects more than one kind of data.

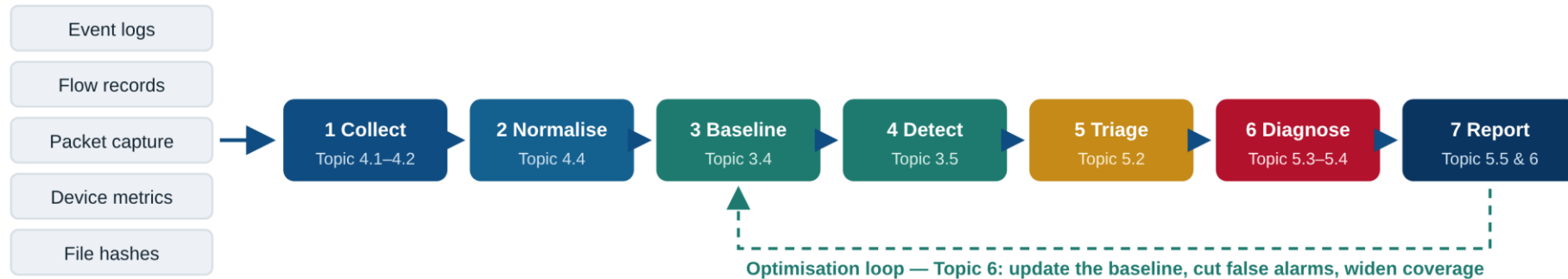


From a raw observation to a report someone can act on

Seven stages. The course topic that teaches each stage is printed under it.

From raw observation to a report someone can act on

Sources



Two properties of this pipeline you are graded on

Stage 3 cannot be skipped: without a baseline, stage 4 has nothing to compare against and every rule becomes a guess.

Stage 7 feeds back into stage 3. A monitoring setup that is never re-baselined degrades into confident, incorrect alerting.



The seven stages (1 of 2)

1

Collect

Gather raw material from the sources you chose: event logs, captured packets, flow records, device metrics, file hashes. Nothing downstream can recover data you never collected.

[Topic 4.1 – 4.2](#)

2

Normalise

Put everything into one consistent form — one time zone, one timestamp format, one field name for ‘source address’. Without this, two tools describing the same event look like two events.

[Topic 4.4](#)

3

Baseline

Record what normal looks like over a representative period, as numbers rather than impressions. This is the reference database the competency names in element 3.4.

[Topic 3.4](#)

4

Detect

Compare live observation against the baseline and against known-bad patterns, and raise an alert when the comparison fails.

[Topic 3.5](#)

The seven stages (2 of 2)

5

Triage

Sort the alerts. Decide which are worth a person's attention, in what order, and which are noise the setup should stop producing.

[Topic 5.2](#)

6

Diagnose

Establish what actually happened and why, using more than one source of evidence. This is the stage the competency grades most heavily.

[Topic 5.3 – 5.4](#)

7

Report and optimise

Write it so the person who must act can act, then feed what you learned back into the baseline, the alert rules and the coverage.

[Topic 5.5 and Topic 6](#)

Detection is not diagnosis, and diagnosis is not a report.

An answer that stops at 'the tool alerted' is an incomplete answer in this course and an incomplete piece of work in the profession. Element 4 of EZ01 asks for an accurate diagnosis of incidents.



Where the data comes from

Five kinds of evidence. Each answers a different question, and no single one is sufficient.

Event logs

Windows Event Log and Linux syslog or journald.
Answers: who authenticated, what service started or stopped, what changed.

Packet capture

A full copy of the traffic, stored as a PCAP file.
Answers: exactly what crossed the wire. Expensive in disk and in privacy.

Flow records

NetFlow and IPFIX.
Answers: who talked to whom, for how long, how much. Cheap to store, no content.

Device metrics

SNMP counters and agent metrics. Answers: how hard is this device working, is it healthy, is a queue growing.

File hashes

SHA-256 over files that should not change.
Answers: has this been altered since the last check.

Data about a network is data about people

A packet capture records what an identifiable person typed, read and sent. Treat every capture, log file, baseline and alert record as personal information: collect the minimum, keep it in the laboratory, and delete it when the exercise is marked.

Québec's Law 25 and section 184 of the Criminal Code both apply. Capturing on a network you share with other people can constitute interception.



The baseline — measuring ‘normal’ as a number

Element 3.4 of the competency calls it the reference database. In practice it starts as two numbers and a threshold.

 μ

the mean — the centre
of normal behaviour

 σ

the standard deviation — how
far
normal usually sits from that
centre

 $\mu + 3\sigma$

the detection threshold —
about 99.7 %
of normal observations fall
below it

Worked on real numbers tonight

168 hourly counts of failed authentications from the
Atelier file server, one representative week.

$\mu = 4.17$ $\sigma = 1.28$ **threshold**
= 8.01

Choose a representative period

A week that contains a holiday, an outage or a
migration produces a baseline that describes
nothing.

Baselines go stale

New staff, new software, a new backup schedule —
all move μ . An un-refreshed baseline alerts on
ordinary change.

Three σ is a starting point, not a law

It assumes roughly symmetric data. On counts that
are heavily skewed, use a percentile instead and
say so.

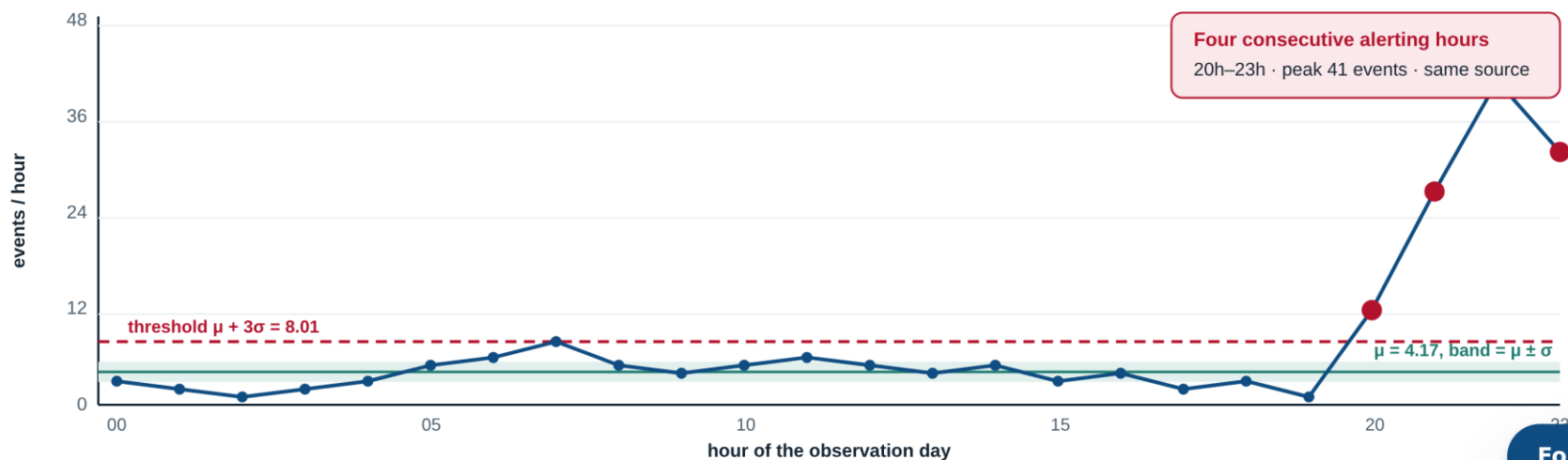


The baseline against a real day

Failed authentications per hour on AN-FS01. The band is $\mu \pm \sigma$, the dashed line is the threshold.

Failed authentications per hour — Atelier Nord-Ouest file server

Baseline from 168 hourly counts: $\mu = 4.17$, $\sigma = 1.28$, threshold $\mu + 3\sigma = 8.01$



Consecutive alerts matter more than a single high hour: one spike is often a mistyped password on a scheduled task; a rising run outside business hours is a pattern.

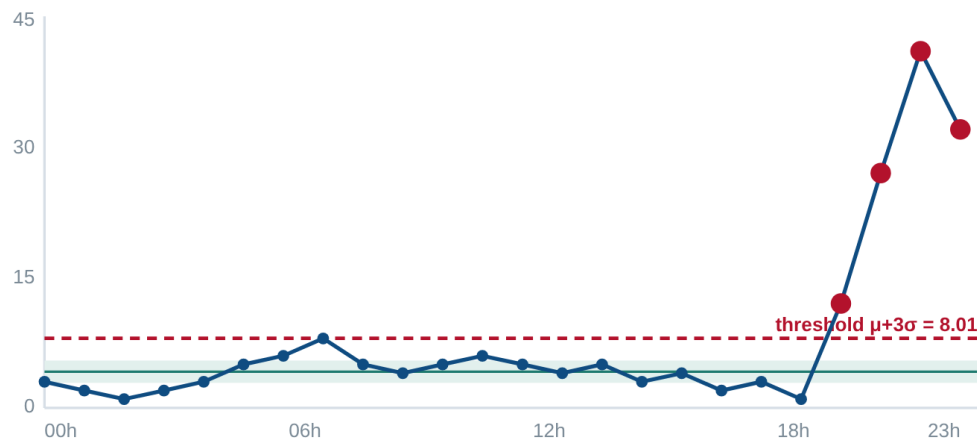
Fold 1..



Detecting it, hour by hour

Thirty lines of Python containing the whole pipeline: collect, normalise, baseline, detect.

Failed authentications per hour — observation day



```
threshold=8.0074
```

```
> column labels written: hour, count, over_threshold  
> 24 observation rows written, 4 of them marked yes  
> Observation sheet written to Baseline_observation_sheet.csv
```

What to watch for while it runs

- The three reference lines are drawn first, from the baseline week. They never move during the run — that is what makes them a reference.
- Each hour is then plotted and compared. Navy means below the threshold, crimson means above it.
- Alerts are collected in a list rather than acted on one at a time. That is what makes triage possible: four consecutive hours mean something different from four scattered ones.
- The run ends by writing the observation sheet — deliverable D3, handed in with Laboratory 2 on 3 October.

Result: **4 alerting hours, first at 20:00, peak 41.**



How a monitoring setup is judged

Five terms you will be expected to use precisely, in laboratories and in both examinations.

True positive

TP

An alert that corresponds to something real. What the setup exists to produce.

False positive

FP

An alert with no real event behind it. Cheap once, ruinous in volume — analysts stop reading.

False negative

FN

A real event that produced no alert. The dangerous failure, because nothing tells you it happened.

MTTD

Mean time to detect — how long between the event starting and the alert being raised.

MTTR

Mean time to respond — how long between the alert and the situation being handled.

The trade-off you cannot escape. Lowering a threshold catches more real events and produces more false alarms. Raising it does the opposite. Element 5 of the competency asks you to improve the quality and accuracy of results — which means tuning this trade-off deliberately, and writing down why.



Who says what good monitoring looks like

You are not inventing this. These are the published references the profession works from.

NIST SP 800-137

Information Security Continuous Monitoring (ISCM). The reference document for what continuous monitoring means and how a programme is structured.

NIST Cybersecurity Framework

Organises security work into Govern, Identify, Protect, Detect, Respond and Recover. This course lives in Detect.

CIS Controls

A prioritised list of safeguards. Controls 8 and 13 — audit log management and network monitoring — are this course.

MITRE ATT&CK

A catalogue of observed adversary techniques with the data sources that reveal each one. Useful when deciding what to collect.

Canadian Centre for Cyber Security

Canada's national technical authority. Publishes advisories and baseline guidance for Canadian organisations.

Québec Law 25 and the CAI

The privacy rules that apply to what monitoring collects, and the Commission that enforces them.

Every one of these is free to read, and all six are linked at the end of the lecture page. **Citing a published standard is what distinguishes a recommendation from an opinion.**



Watch for Lesson 1.3

Watch before the next class — each under 15 minutes

Professor Messer — the CIA triad · 5 min <https://www.youtube.com/watch?v=SBcDGB9l6yo>

Learn Wireshark in ten minutes · a first look at packet capture <https://www.youtube.com/watch?v=lb1Dw0elw0Q>

Before you open Wireshark on anything

- Install it inside the laboratory virtual machine, not on your own operating system.
- Capture only on an interface of a machine you own or that the exercise designates as a target.
- Never enable promiscuous or monitor mode on a network you share with other people. That records their communications and may constitute interception under section 184 of the Criminal Code.
- Delete captures when the exercise is marked. Section 6.7 of the Cyber.SoHo notice requires it, and so does Law 25.



LESSON 4 OF 4

1.4 The laboratory environment

The isolated environment every exercise in this course runs inside, why it must stay isolated, and how to prove that it is.



The Authorized Lab Environment

A defined term from the Cyber.SoHo notice. Three conditions, all of which must hold at once.

1

You own it, or you are permitted to use it for the exercise

Your own computer, or cloud or rented resources you are entitled to use for this purpose, in compliance with the provider's terms.

2

It is disconnected from, or securely segregated from, every other network

No production network, no institutional network, no corporate network, no public network and no third party's network.

3

It contains no real personal data and no system belonging to anyone who has not consented in writing

Fictional organisations, fictional users, synthetic data. Never an employer's configuration, never a family member's machine.

Designated Target. The system a laboratory guide expressly names as the thing you are to work on. Nothing becomes a designated target because it is convenient, because it is reachable, or because it resembles one.

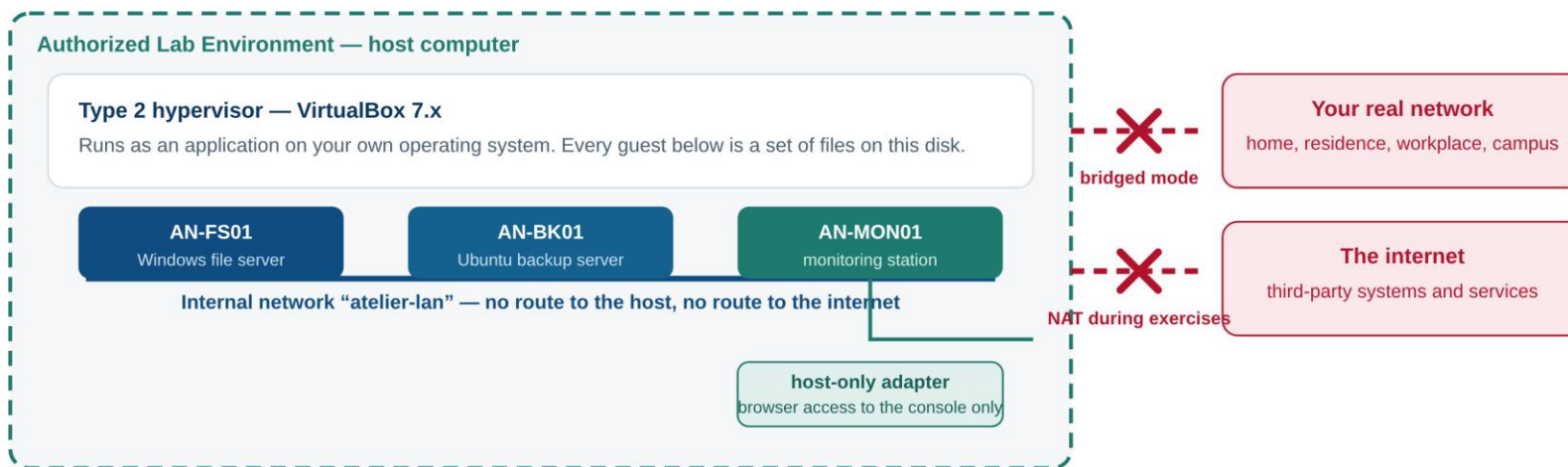


The laboratory, and the boundary that defines it

Everything inside the dashed boundary is the Authorized Lab Environment. Nothing crosses it except by an instruction in a laboratory guide.

The laboratory, and the boundary that defines it

Everything inside the dashed boundary is the Authorized Lab Environment. Nothing crosses it except by an instruction in a laboratory guide.



The single most common failure in this course

An adapter left in bridged mode after a download session. Run the pre-flight check before every exercise, not once at the star



The five network modes, and which you may use

The single setting that decides whether your laboratory is isolated or not.

MODE	REACHES THE INTERNET	REACHES OTHER LAB MACHINES	EXPOSED TO YOUR REAL NETWORK	USE IN THIS COURSE
NAT	yes	no	no inbound	Only to download updates, then off
NAT Network	yes	yes	no inbound	Only when an exercise says so
Bridged adapter	yes	yes	YES, fully	PROHIBITED
Host-only adapter	no	yes	no	Permitted
Internal network	no	yes	no	Preferred for all exercises

The one failure that happens every term. A student switches an adapter to NAT or bridged to download a package, finishes, and forgets to switch it back. The machine then sits on the real network while the exercise runs. Check before every exercise, not once at the start of term.



What to build before Laboratory 1

Three guest machines and one host. Build them this week — not on the evening of 25 September.

AN-FS01

Windows Server, evaluation edition

File server holding the client drawings. The source of the event logs you will baseline.

2048 MB · 40 GB · internal network

AN-BK01

Ubuntu Server LTS

Backup server. The machine whose silent failure is an availability incident.

1024 MB · 20 GB · internal network

AN-MON01

Ubuntu Server LTS

The monitoring station. Your tools run here, and nothing else does.

2048 MB · 40 GB · internal + host-only

Host computer

64-bit processor with virtualisation enabled in firmware · 8 GB memory minimum · 80 GB free

Hypervisor

Oracle VirtualBox 7.x, downloaded from [virtualbox.org](https://www.virtualbox.org) and nowhere else

Snapshot

Take one on every machine before every exercise. Section 6.4 of the notice requires it

Verify what you download

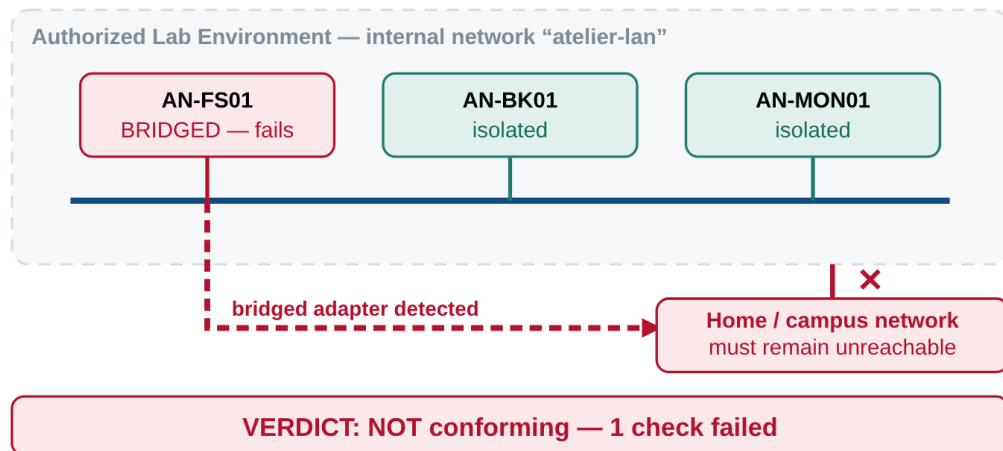
Check the publisher's SHA-256 against your file, and check the installer with VirusTotal



Proving the laboratory is isolated

A pre-flight check you run before every exercise. Tonight it deliberately fails.

Isolation status, checked machine by machine



then run this script again.

```
> Remediation: Settings > Network > Adapter 2 > Attached to:  
Internal Network  
> exit 1 (a non-zero status means: do not start the exercise)
```

Four checks, in order

- Is any adapter in bridged mode? AN-FS01 fails here — an adapter was left bridged after a download session.
- Does every machine have a snapshot to restore to?
- Is there enough free disk for captures and log stores?
- Is a signed Annex A present for this course?

Verdict: **NOT conforming. One check failed, so the script exits non-zero and the exercise does not begin.**

A PowerShell version is provided for Windows hosts. Both produce deliverable D4.



Watch and install for Lesson 1.4

Watch before the next class — each under 15 minutes

Install Ubuntu in VirtualBox — beginner guide · 11 min <https://www.youtube.com/watch?v=Hm87A8R3hIU>

VirtualBox network modes explained · 9 min <https://www.youtube.com/watch?v=lcneGkEYIoQ>

Download only from the publisher

- virtualbox.org — the hypervisor
- ubuntu.com/download/server — the Linux guests
- wireshark.org — the packet analyser, from Topic 4

Never a mirror, a 'download manager' or a search advertisement.

Check before you run

- virustotal.com — submit the URL, and the file or its hash
- Google Safe Browsing — check the domain's reputation
- Compare the publisher's published SHA-256 with your download

A clean result is never a guarantee. A mismatch is a definite answer.





The four deliverables and where they land

Nothing produced tonight is throwaway work. Each item is handed in with a graded laboratory.

D1**Monitoring scope register**

The eight assets, why each is monitored, which elements of EZ01 it demonstrates, what evidence you hand in.

Laboratory 1 · 25 September**D2****Delivery and study plan**

All nine graded items in your own calendar, with review points before each.

Laboratory 1 · 25 September**D3****Baseline observation sheet**

μ , σ and the threshold, plus the twenty-four observed hours marked over or under.

Laboratory 2 · 3 October, reused in Laboratory 3**D4****Lab conformity report**

The output of the pre-flight check, showing the environment was isolated when you worked.

Attached to Laboratories 1, 2 and 3

All four start from files you can download tonight from the lecture page — the scope register, the schedule, the observation sheet and the two pre-flight scripts are all there, already filled with the case-study data.





CLOSING

Before Thursday 17 September

Six things. None of them takes long, and all of them are cheaper now than later.



Read the Cyber.SoHo notice and sign Annex A

Section 7.2 applies to this course. Laboratory access requires the signed page.



Put all nine dated items in your calendar

With an alert three days before each. Block 1 October and 19 October for travel.



Open a free BANQ account

O'Reilly and LinkedIn Learning, at no cost, for the whole programme.



Install VirtualBox and build the three machines

From the publisher's site only. Verify the SHA-256 before you run the installer.



Set every adapter to internal network and snapshot

Then run the pre-flight check and keep the output — that is deliverable D4.



Watch the two videos for Lesson 1.1

Twelve minutes and four minutes. They make the first laboratory shorter.





Everything linked tonight

All of these are official publisher or institutional sites. The lecture page carries the full directory with a note on each.

Course governance and law

- Cyber.SoHo Legal Notice v2.0
- Dawson College ISEP
- Criminal Code (laws-lois.justice.gc.ca)
- Law 25 — P-39.1 (LégisQuébec)
- Commission d'accès à l'information
- BANQ registration

Standards and references

- NIST SP 800-137
- NIST Cybersecurity Framework
- CIS Controls
- MITRE ATT&CK
- Canadian Centre for Cyber Security
- RFC Editor

Software and safety checking

- VirtualBox
- Ubuntu Server
- Wireshark
- Zeek, Suricata, Security Onion
- VirusTotal — URL and file
- Google Safe Browsing

Before you open any link that is not on this list: read the domain letter by letter, check it with VirusTotal and Google Safe Browsing, and prefer the publisher's own site over any mirror. Annex C of the notice is the full checklist.



NEXT CLASS · THURSDAY 17 SEPTEMBER, 18:00, ONLINE

Topic 1 — Foundations of cybersecurity

- 1.1 What cybersecurity is and why organisations need it
- 1.2 The CIA triad in detail, with worked examples



Questions between classes: MIO, or office 6C.9

Asen Grozdanov · Cyber.SoHo · Montréal, Québec

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